MIL-STD-00410C (MR)
20 October 1972
USED IN LIEU OF
MIL-STD-410B
3 May 1972

MILITARY STANDARD

QUALIFICATION OF INSPECTION PERSONNEL

(Magnetic Particle and Penetrant)

FSC MISC

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Army Materials and Mechanics Research Center
Watertown, Massachusetts 02172

Qualification of Inspection
Personnel (Magnetic
Particle and Penetrant)

MIL-STD-00410C(MR)

1. This limited coordination Military Standard has been prepared by the Army Materials and Mechanics Research Center based upon currently available information, but it has not been approved for promulgation as a coordinated revision of MIL-STD-410. It is subject to modification. However, pending its promulgation as a coordinated Military Standard, it is mandatory for use by the Department of the Army.

2. Recommended corrections, additions or deletions should be addressed to the Director, Army Materials and Mechanics Research Center, Watertown, Massachusetts 02172.

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QUALIFICATION OF INSPECTION PERSONNEL
(Magnetic Particle and Penetrant)

1. SCOPE

1.1 Scope. This standard covers requirements and tests for qualification of personnel to inspect materials and parts by the magnetic particle and penetrant inspection methods.

2. REFERENCED DOCUMENTS

2.1 Government documents. The issues of the following documents, in effect on date of invitation for bids, form a part of this standard to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-6866 - Inspection, Penetrant Method of
MIL-I-6868 - Inspection Process; Magnetic Particle
MIL-M-11472 - Magnetic Particle Inspection; Process, for Ferromagnetic Materials

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING

Recommend Practice No. SNT-TC-1A, Nondestructive Testing Personnel Qualification and Certification: Supplement B, Magnetic Particle Method; and Supplement D, Liquid Penetrant Testing Method.

Nondestructive Testing Handbook

(Application for copies should be addressed to the American Society for Nondestructive Testing, 914 Chicago Avenue, Evanston, Illinois 60202)

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MAGNAFLUX CORPORATION

Principles of Magnetic Particle Inspection

(Application for copies should be addressed to the Magnaflux Corporation, 7300 West Lawrence Avenue, Chicago, Illinois 60656)

3. GENERAL REQUIREMENTS

3.1 Policy. Inspection personnel shall be qualified in accordance with this standard before conducting magnetic particle and penetrant inspection.

3.2 Responsibility. The supplier (employer of the inspection personnel) is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in this standard when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

3.3 Government Specifications. The procedures, materials, equipment, and other details for qualification for magnetic particle and penetrant inspection shall be in accordance with Specifications MIL-I-6866 and MIL-I-6868 as applicable.

3.3.1 Applicability of qualifications. Qualification in accordance with this standard is applicable to fluorescent and nonfluorescent magnetic and penetrant inspection methods.

3.4 Qualification Examinations. Inspection personnel shall be qualified by examinations, which can be oral and operational, written and operational, or a combination of all three.

3.5 Maintenance of Qualification. Qualified status of an inspector shall be maintained at the option of the employer by (a) a continuing record of satisfactory workmanship in accordance with 4.2.1, or (b) by requalification every 3 years in accordance with requirements listed herein.

3.6 Revocation of Qualification. Evidence that either inspection or evaluation of indications is not being accomplished with prescribed requirements shall be sufficient cause for revocation of qualification. Requalification of any inspector shall be required whenever there is valid reason to question the proficiency of the inspector or quality of workmanship.

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4. DETAIL REQUIREMENTS

4.1 Interpretation of Indications. The interpretation of indications with the aid of established standards is the prime responsibility of the magnetic particle or penetrant inspector.

4.2 Record of Qualification. After the applicant has successfully completed all tests, the employer shall complete and retain the essential information shown in Figure 1.

4.2.1 Continuing record of workmanship. A continuing record of workmanship shall be based on a statistical reinspection plan of the material previously accepted or rejected by the inspector. The reinspection shall be performed by a different qualified inspector.

4.3 Requirements for Qualification

4.3.1 Equipment and materials. The applicant shall possess a working knowledge of the following equipment and processes.

a. For magnetic particle inspection, the operation of equipment, application of the materials, operation of demagnetization equipment, and operation of accessory equipment used in connection with the above; the function of the materials used in the wet and dry inspection procedures and the magnetic substance properties, such as magnetic characteristics and color, that are significant in the magnetic particle inspection methods.

b. For penetrant inspection, the function of penetrants and characteristics of developers, including the procedures and the required tests outlined in Specification MIL-I-6866.

c. Procedures and requirements covered by the applicable documents listed in 2.1.

4.3.2 Inspection procedures. The applicant shall possess a thorough knowledge of the following conditions:

a. Cleaning and any other necessary preparation of parts for inspection.

b. Methods of applying penetrants and developers (penetrant inspection only).

c. The application of the continuous and residual magnetization procedures, including proper orientation of parts in solenoids or between contact points to accomplish longitudinal or circular magnetization (magnetic particle inspection only).

d. Methods of applying the magnetic substance for the wet continuous and wet or dry residual magnetization procedures (magnetic particle inspection only).

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RECORD OF QUALIFICATION

MAGNETIC PARTICLE AND PENETRANT INSPECTION PERSONNEL

Date_____

Name of applicant_____

Address_____

Name of employer_____

Address_____

_____

Type of inspection - magnetic particle or penetrant, or both_____

_____

EXPERIENCE

Employer Years Type of work

_____

TECHNICAL TRAINING

_____

Applicant_____
(Signature)

VISION TEST

	Distant vision		Near Vision	
Uncorrected	RE_____	Uncorrected	RE_____	
	LE_____		LE_____	
Corrected	RE_____	Corrected	RE_____	
	LE_____		LE_____	
Color Vision	_____	Date of Test	_____	
Examiner	_____	Remarks	_____	

EXAMINATION

Oral_____Written_____Raw score_____Equipment and process_____
(if applicable)

Employer representative_____
(Signature)

FIGURE 1. Suggested form for record of qualification

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a. For magnetic particle inspection, the operation of equipment, application of the materials, operation of demagnetization equipment, and operation of accessory equipment used in connection with the above; the function of the materials used in the wet and dry inspection procedures and the magnetic substance properties, such as magnetic characteristics and color, that are significant in the magnetic particle inspection methods.

b. For penetrant inspection, the function of penetrants and characteristics of developers, including the procedures and the required tests outlined in Specification MIL-I-6866.

c. Procedures and requirements covered by the applicable documents listed in 2.1.

4.3.2 Inspection procedures. The applicant shall possess a thorough knowledge of the following conditions:

a. Cleaning and any other necessary preparation of parts for inspection.

b. Methods of applying penetrants and developers (penetrant inspection only).

c. The application of the continuous and residual magnetization procedures, including proper orientation of parts in solenoids or between contact points to accomplish longitudinal or circular magnetization (magnetic particle inspection only).

d. Methods of applying the magnetic substance for the wet continuous and wet or dry residual magnetization procedures (magnetic particle inspection only).

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RECORD OF QUALIFICATION

MAGNETIC PARTICLE AND PENETRANT INSPECTION PERSONNEL

Date_____

Name of applicant_____

Address_____

Name of employer_____

Address_____

_____

Type of inspection - magnetic particle or penetrant, or both_____

_____

EXPERIENCE

Employer Years Type of work

_____

TECHNICAL TRAINING

_____

Applicant_____
(Signature)

VISION TEST

	Distant vision		Near Vision	
Uncorrected	RE		RE	
	LE		LE	
Corrected	RE		RE	
	LE		LE	
Color Vision			Date of Test	
Examiner			Remarks	

EXAMINATION

Oral_____ Written_____ Raw score_____ Equipment and process_____
(if applicable)

Employer representative_____
(Signature)

FIGURE 1. Suggested form for record of qualification

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e. Methods of demagnetization and tests for residual magnetism after demagnetizing treatment (magnetic particle inspection only).

f. Cleaning and treatments of parts or assemblies after inspection.

g. Identification or marking of parts after inspection.

4.3.3 Technical requirements.

a. General theory, operating principles, and characteristics of material used in magnetic particle and penetrant inspection.

b. Magnetic particle and penetrant inspection requirements contained in Specifications MIL-I-6866, MIL-I-6868 and MIL-M-11472.

c. Interpretation of magnetic particles and penetrant inspection indications upon which to base acceptance or rejection of parts in accordance with applicable established standards, specifications, or other mandatory publications.

d. Stages of manufacture or processing when magnetic particle and penetrant inspection are applicable in accordance with service specifications.

4.4 Vision Tests. Vision tests shall be performed by an oculist or optometrist or by any other professionally recognized personnel. One year from the effective date of qualification, and each year thereafter, qualified personnel shall be required to pass the vision tests specified herein.

4.4.1 Vision requirements. The following minimum vision requirements shall apply:

a. Distant vision shall equal 20/30 in at least one eye, either uncorrected or corrected.

b. Near vision shall be equal to normal vision, as prescribed by the optical profession, in at least one eye, either corrected or uncorrected.

c. The personnel shall be capable of distinguishing and differentiating between colors used in the process involved.

4.5 Written and oral examinations. Applicants shall take written or oral tests, or both, based on governmental documents listed in 4.3.3b and the nongovernmental publications listed in 2.2. The tests will cover topics such as listed below:

a. Operation of magnetic particle and penetrant inspection equipment.

b. Methods of magnetization (when applicable).

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c. Common types of discontinuities causing indications, irrelevant indications and indications generally considered acceptable.

d. Application of various types of penetrants and developers.

e. Cleaning, demagnetization (when applicable), marking, and stamping procedures.

4.5.1 Additional written or oral examinations. Additional written or oral examinations shall include the following subjects:

a. The interpretation and application of inspection requirements in Government specifications and other mandatory publications, which are applicable to the inspection performed in the facility.

b. The interpretation of indications and the acceptance or rejection of parts in accordance with specifications, standards, or other mandatory publications applicable to inspection in the facility.

4.6 Operational Tests. Operational tests shall include tests for the control or inspection media and complete inspection of parts or assemblies containing acceptable, borderline rejectable indications, and representative of the work processed in the facility for which qualification will be acceptable.

4.7 Alternate Qualification. Qualification of personnel in accordance with SNT-TC-1A Supplement B and Supplement D will be recognized as qualifying personnel to this standard providing all other conditions of this document are met and are duly documented.

Custodians:
Army - MR

Preparing activity:
Army - MR

Project Number MISC-A868

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Army Materials and Mechanics Research Center
Watertown, Massachusetts 02172

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-STD-00410C(MR), Qualification of Inspection Personnel (Magnetic Particle and Penetrant)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

1

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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